Discrete Math Midterm Solutions

1. **T(rue) or F(alse).** You don't have to give a reason, but it might get you half a point if you are wrong.

   (a) For real numbers x and y:

   $$\lceil (\lfloor x \rfloor + \lfloor y \rfloor) \rceil = \lfloor x \rfloor + \lfloor y \rfloor.$$

   Solution

   T

   (b) A function $f : X \to X$ is one-to-one if and only if it is onto.

   Solution

   F: Need not be true when X is infinite.

   (c) Every minimal element of a partial order is a minimum element.

   Solution

   F: Other way round.

   (d) The set $[n] = \{1, 2, \ldots, n\}$ has exactly $n/2$ subsets of odd size.

   Solution

   F: It has 2^{n-1}.

   (e) When $m \leq n$, there are $\frac{n!}{(n-m)!}$ injective functions from $[m]$ to $[n]$?

   Solution

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   (f) $n^2 = O(n^2 \ln n)$.

   Solution

   T

2. How many permutations of $[n]$ have exactly one cycle?

Solution

To build a permutation σ with one cycle, so in cycle notation looks like

$$\sigma = (a_0\, a_1\, \ldots\, a_{n-1})$$

we can choose a_0 to be 1, and then when choosing a_i for $i = 1 \ldots, n-1$ we can choose any of the $n-i$ elements not already chosen. So there are $\boxed{(n-1)!}$ such permutations.

Solution

Alternately you might observe that there are $n!$ ways to choose the a_i, and each cyclic permuation as n different representations: $(a_0\, \ldots\, a_{n-1}), (a_1\, \ldots\, a_{n-1}\, a_0)$... so on. So there are $(n-1)! = \frac{n!}{n}$ such permuations.

3. What is the smallest $p \geq 1$ such that σ^p is the identity permutation, when σ, in cycle notation, is

$$\sigma = (1\,2\,3)(4\,5)(6\,7\,8\,9)(10)(11\,12\,13\,14\,15)?$$

Solution

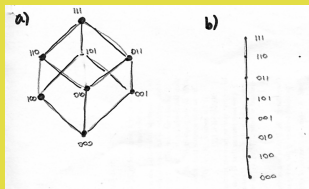
For an element such as 1 is a cycle $(1\,2\,3)$ of length 3, we have that

$$(\phi^1(1), \phi^2(1), \phi^3(1), \phi^4(1), \phi^5(1), \phi^5(1)...) = (2, 3, 1, 2, 3, 1, ...).$$

Indeed, ϕ^p takes an element i to itself when p is a multiple of the length of the cycle that i is in. Thus ϕ^p takes every element to itself if p is a multiple of all the lengths $3, 2, 4, 1, 5$. The p we want is thus the lcm of these numbers: $\boxed{60}$

4. How many linear extensions are there of the poset $\mathcal{B}_3$ (the Boolean lattice)? (For partial marks, draw the Hasse Diagram of the poset $\mathcal{B}_3$ and of a linear extension of it.)

Solution



There are $36 + 12 = \boxed{48}$ linear extensions of $\mathcal{B}_3$.

Here is why: Let $f : \mathcal{B} \to (\{0, 1, \ldots, 7\}, \leq)$ be a linear extension of $\mathcal{B}_3$. Then clearly $f(000) = 0$ and $f(111) = 7$. Any choice of $1, 2, 3$ for $f(001)$, $f(010)$ and $f(100)$ (along with any choice of $4, 5, 6$ for $f(011)$, $f(101)$ and $f(110)$) is allowed. There are 36 such choices. But these are not the only linear extensions. We can have, say $f(001) > f(110)$. If $f(001) > 3$ then we must have: $f(001) = 4$, $f(110) = 3$, and $f(011) > 4$, $f(101) > 4$, $f(100) < 3$ and $f(010) < 3$. There are 4 ways to choose these values. Similarily for each of $f(100) > 3$ and $f(010) > 3$ we have 4 linear extensions.

5. Determine the number of ordered pairs (A, B) with

$$A \subseteq B \subseteq [n].$$

Solution

For such a pair, each $i \in [n]$ can be
(a) in neither set,
(b) in just B, or
(c) in A and B.
As there are 3 choices for each i, there are 3^n such ordered pairs.

6. Among the numbers in $[10^{10}]$ are there more that have 9 as a digit, or more that don't?

Solution

Surprisingly, $\boxed{\text{more that do}}$. Let $X = [10^{10}]$ and $N \subset X$ be the numbers that don't have a 9 as a digit. Counting $|N|$ is simplified by removing the number 10000000000 and adding the number 0000000000, this doesn't change its size. We can now choose a number in N by placing any of the other 9 digits in each of 10 places, so $\boxed{|N| = 9^{10}}$. (Some people made this much more difficult, counting $|X \setminus N|$ using PIE.)
Now, we must show that $|N| < |X \setminus N|$ or equivalently that N has less than have the numbers $|N|/|X| < 1/2$:

$$\frac{|N|}{|X|} = \left(\frac{9}{10}\right)^{10} = (1 - \frac{1}{10})^{10} < (e^{\frac{-1}{10}})^{10} = e^{-1} < 1/2.7 < .4 < 1/2.$$

(This was what I meant for the question to be about, but it turned out it was a bit difficult. Sorry. So I only count this for one point.) If using e^{-1} scares you, we can use the binomial theorem instead, and show $\frac{|X|}{|N|} > 2$:

$$\begin{aligned} \frac{|X|}{|N|} &= \left(\frac{10}{9}\right)^{10} = (1.1111...)10 > (1 + .1)^{10} \\ &= 1 + \binom{10}{2}(.1) + \binom{10}{2}(.1)^2 + ... > 1 + 1 + .4 \end{aligned}$$

7. How many numbers in [100] are not divisible by a square of an integer greater than 1?

Solution

Let $X = [100]$ and for $i = 2, 3, 5, 7$ let A_i be the multiples of i^2 in X. For $i = 6, 10$ let B_i be the multiples of i^2 in X; and observe that $B_6 = A_2 \cap A_3$, and $B_{10} = A_2 \cap A_5$, are the only intersections of more than one of the A_i that are non-empty. The dual form of PIE tells us that the number of elements in none of the A_i is

$$\begin{aligned} &|X| - |A_2| - |A_3| - |A_5| - |A_7| + |B_6| + |B_{10}| \\ &= 100 - (25 + 11 + 4 + 2) + (16 + 10) = \boxed{61}. \end{aligned}$$

Proofs

Do **two** of the following problems. (Use back of page if you need room.)

8. Show that every finite partial order can be embedded into $(N, |)$.

Solution

Let $(P, \leq)$ be a finite partial ordering. We may rename the elements of P without changing anything, assume therefore that P is a set of distinct prime numbers (their ordering as numbers is unrelated to the ordering $\leq$).

For each $p \in P$ let $D(p) = \{q \in P \mid q \leq p\}$ be its downset; we claim that $f : P \to N$ defined by $f(p) = \prod_{q \in D(p)} q$ is an embedding of $(P, \leq)$ into $(N, |)$. To see that f is injective, assume that $f(p) = f(q)$. Then $p \in D(p) = D(q)$ and so $p \in D(q)$, meaning that $p \leq q$. But the same argument gives that $q \in D(p)$ and so $q \leq p$. Thus $p = q$ by the antisymmetry of P, and f is an injection.

Now let $p \leq q$. Then $D(p) \subset D(q)$ and so $f(p) \mid f(q)$. On the other hand let $f(p) \mid f(q)$. As $p \mid f(p)$ we have that $p \mid f(q)$ and so $p \in D(q)$, thus $p \leq q$. Thus $p \leq q \iff f(p) \mid f(q)$ showing that f is an embedding.

9. Prove that when $r \leq n$ we have

$$\binom{r}{r} + \binom{r+1}{r} + \cdots + \binom{n}{r} = \binom{n+1}{r+1}.$$

Solution

The most straight-forward way is to use induction on $n \geq r$ for fixed r, using the identity $\binom{n+1}{r+1} = \binom{n}{r+1} + \binom{n}{r}$ for the induction step.

Using $n = 1$ as the base step and inducing on n will work, but you have to be careful about the induction step. If you claim that you prove the result for some n, this means that the identity holds for all $r \leq n$. When you do your induction step, you get that for the $n+1$ case, the identity holds for all $r \leq n$, so you still have to prove the case $r = n+1$.

Some people tried a base case of $r = 1$, as the identity then holds for all $n \geq r$, but nobody managed to sucessfully pull off an induction on r.

Solution

Some people used a 'recursive proof' by successively appling the identity $\binom{a}{b} = \binom{a-1}{b-1} + \binom{a-1}{b}$ to the last term:

$$\begin{aligned}\binom{n+1}{r+1} &= \binom{n}{r} + \binom{n}{r+1} \\ &= \binom{n}{r} + \binom{n-1}{r} + \binom{n-1}{r+1} \\ &= \binom{n}{r} + \binom{n-1}{r} + \binom{n-2}{r} + \binom{n-2}{r+1}\end{aligned}$$

This works, but you have to be careful at the last step. You get to

$$\binom{n+1}{r+1} = \binom{n}{r} + \binom{n-1}{r} + \binom{n-2}{r} + \cdots + \binom{r+1}{r} + \binom{r+1}{r+1}.$$

This is not the expression you want, and the identity doesn't work anymore. You must now observe that $\binom{r+1}{r+1} = 1 = \binom{r}{r}$ to get the identity that you want.

Solution

One student gave a beautiful combinatorial proof, observing that to choose a set X of $r+1$ elements from the set $[n+1]$ element set. On the one hand there are clearly $\binom{n+1}{r+1}$ ways to choose the set. However, one can first choose the maximum element M of X, and then choose the other n elements in $\binom{M}{n}$ ways. The maximum element M must be in $\{r+1, r+2, \ldots, n+1\}$ so summing the choices over the different values of M we can choose X in

$$\sum_{M=r+1}^{n+1} \binom{M-1}{r} = \binom{r}{r} + \binom{r+1}{r} + \ldots \binom{n+1}{r}$$

ways. Thus this sum is $\binom{n+1}{r+1}$.

10. Show that if G has a uv-walk, then it has a uv-path.

Solution

Assume that G has a uv-walk, and that the length ℓ walk $W = v_0, v_1, \ldots, v_\ell$ is a shortest such walk. We show that W is in fact a uv-path. Towards contradiction, assume it is not. Then there are indices $i < j$ such that $v_i = v_j$. But then $W' = v_0, v_1, \ldots, v_i, v_{j+1}, \ldots, v_\ell$, (or $W' = v_0, v_1, \ldots, v_{i-1}, v_j$ in the case that $j = \ell$) is a uv-walk of length $\ell - (j - i)$. This is shorter than W, so contradicts our choice of W as the shortest uv-walk.